Dmitrii **Altukhov**

Machine Learning Engineer · PhD (Computer Science)

Amsterdam, The Netherlands

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Profile

Machine Learning Engineer with a strong software engineering background and track record of developing high-performance algorithms and scalable production systems. PhD in Computer Science with expertise in numerical optimization and translating research into production-ready software.

Programming languages Python, C/C++, Java, Javascript (React, NodeJS)

Quantitative skills Algorithms and data structures, numerical optimization, machine learning, linear algebra, statistics

Tools & Technologies PyTorch, Docker, Git, CI/CD, AWS, Kafka, Redis, SQL, Kubernetes

Experience

Creative Fabrica — *Amsterdam, the Netherlands*

Python engineer -> ML engineer -> Senior ML engineer — *Dec. 2023 – present*

- Own the generative AI Python codebase, infrastructure, and internal testing and QA tool.
- Developed advanced algorithm for color quantization that became foundational for industry-leading image vectorization pipeline.
- Designed and maintained distributed event-driven architecture using Kafka, Redis queues, and Kubernetes.
- Redesigned the architecture for image generation backend, significantly improving scalability and system performance.

SBDA Group — *Remote*

Data Scientist (contract) — *Sep. 2023 – Oct 2023*

- Designed and implemented an algorithmic solution in Python for optimizing a manufacturing supply chain process.

Artificial Intelligence Research Institute (AIRI) — *Moscow, Russia*

Research fellow — *Feb. 2022 – Apr. 2023*

- Built and trained deep learning model to predict speech from MEG data in **PyTorch**, meg_speech_decoding on GitHub.
- Developed real-time **Python** pipeline for EEG timeseries prediction using Kalman filtering.

Higher School of Economics, Centre for Cognition and Decision Making — *Moscow, Russia*

Junior research fellow — *Feb. 2015 – Dec. 2021*

- Accelerated various parts of my PhD codebase to enable Monte-Carlo statistics, using **CUDA** (cuBLAS) and **C**.
- Led 4-people group on a project for real-time ML feature extraction and GPU-accelerated 3D visualization of brain activity in **Python**.
- Sped up by a factor of 10 the beamformer inverse solver in MNE-python by modifying the algorithm for vectorized computations.

University of Montreal, CERNEC lab. — *Montreal, Canada*

Visiting researcher, temporary position — *Oct. – Dec. 2015, May 2016 – Dec. 2016*

- Co-developed an open-source **Python** package for GPU-accelerated brain data visualization Visbrain, Combrisson et al. [2019].

Scientific Research Institute of System Analysis — *Moscow, Russia*

Junior research fellow — *Jun. 2011 – Jan. 2015*

- Developed a droplet evaporation model in **C++**; tested **CUDA**-accelerated cluster software for simulations of flow in jet engines.

Education

Ph.D. in Computer Science — *Moscow, Russia*

Higher School of Economics, Faculty of Computer Science — *Jan. 2016 – Nov. 2021*

GPA: 4.0; proposed an algorithmic framework for brain connectivity estimation; research published in the leading domain journals.

Specialist degree in Mechanics *(Masters equivalent)* — *Moscow, Russia*

Lomonosov Moscow State University, Department of Mechanics and Mathematics — *Sep. 2008 – Jun. 2013*

Thesis: “Enhancement and validation of LOGOS software for simulations of the reactive fluid flows.”

Algorithms, Part 1 and Part 2 — *Online*

Coursera, Princeton University — *2020*

Comprehensive Algorithms and Data Structures course in **Java**.